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Diabetic retinopathy screening with confocal fundus camera and artificial intelligence - assisted grading

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Abstract

Purpose: Screening for diabetic retinopathy (DR) by ophthalmologists is costly and labour-intensive. Artificial Intelligence (AI) for automated DR detection could be a clinically and economically alternative. We assessed the performance of a confocal fundus imaging system (DRSplus, Centervue SpA), coupled with an AI algorithm (RetCAD, Thirona B.V.) in a real-world setting.

Methods: 45° non-mydriatic retinal images from 506 patients with diabetes were graded both by an ophthalmologist and by the AI algorithm, according to the International Clinical Diabetic Retinopathy severity scale. Less than moderate retinopathy (DR scores 0, 1) was defined as non-referable, while more severe stages were defined as referable retinopathy. The gradings were then compared both at eye-level and patient-level. Key metrics included sensitivity, specificity all measured with a 95% Confidence Interval.

Results: The percentage of ungradable eyes according to the AI was 2.58%. The performances of the AI algorithm for detecting referable DR were 97.18% sensitivity, 93.73% specificity at eye-level and 98.70% sensitivity and 91.06% specificity at patient-level.

Conclusions: DRSplus paired with RetCAD represents a reliable DR screening solution in a real-world setting. The high sensitivity of the system ensures that almost all patients requiring medical attention for DR are referred to an ophthalmologist for further evaluation.

Keywords

Diabetic retinopathy screening, ophthalmologist referral, artificial intelligence, confocal fundus camera, accuracy study

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Introduction

Diabetic retinopathy(DR)is a chronic and progressive complication of diabetes. If left undetected or untreated, it can lead to vision loss.[1–3] The diagnosis typically involves a careful examination of the retina by an ophthalmologist or a trained diabetologist using various techniques, which might be time-consuming and expensive.

The advent of Artificial Intelligence (AI) has the potential to introduce significant changes in large-scale DR detection by making the screening process faster, more accessible, and affordable.[4,5] Automated analysis of retinal images based on AI (AI-based systems), are able to provide an indication of presence or absence of DR pathology. The key step in building-up such systems is the training of the algorithm, for which thousands of fundus images labelled by experts with DR grading are needed. Once trained, the AI-based system can examine new retinal images and determine DR presence and grading, thus allowing patients' referral.[6] Furthermore, AI doesn't

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suffer from fatigue, ensuring consistent performance. Since AI-based systems are able to process data at a faster rate than human specialists, large-scale screenings can be conducted in a short amount of time, allowing to screen vast populations and thus improving patients' access to care.[4,6]

The frequency and methods for detecting DR with standard screening programs are labor-intensive and based on a patient's individual risk factors for DR, including the type and duration of diabetes, glycemic control, and the presence of other conditions like high blood pressure. On the other hand, screenings built on AI-based systems can significantly lower healthcare costs by reducing the need for specialized human resources and thus making the screening process more efficient.[4] Furthermore, AI-based systems can manage large amounts of data, providing insights for public health initiatives and research.[7]

An example of such AI-based systems is RetCAD, an AI diagnostic software linked with a user-friendly confocal fundus camera. The system analyses images of the retina taken with a fundus camera and provides a result for the presence and a score of DR severity.

When adopting an AI-based system for screening purposes, the major concern is related to its reliability and especially to the risk of missing positive patients. The purpose of this study was to evaluate the performance of a screening system combining DRSplus confocal fundus camera and RetCAD AI software through iCare ILLUME with human grading as gold standard.

Materials and methods

We carried out an observational cross-sectional study in two sites (a diabetes centre and an eye centre) located in ASLTO5, a local health unit in Turin's metropolitan area, Italy. Participants included of adult patients (age above 18 years) with a diagnosis of any diabetes mellitus type, except for gestational diabetes. These individuals were enrolled for DR screening during their regular diabetes check-up between October 2022 and March 2023. A total of 1006 eyes of 506 patients were collected. The study complied with the Declaration of Helsinki and the Good Clinical Practice (GCP) guidelines.

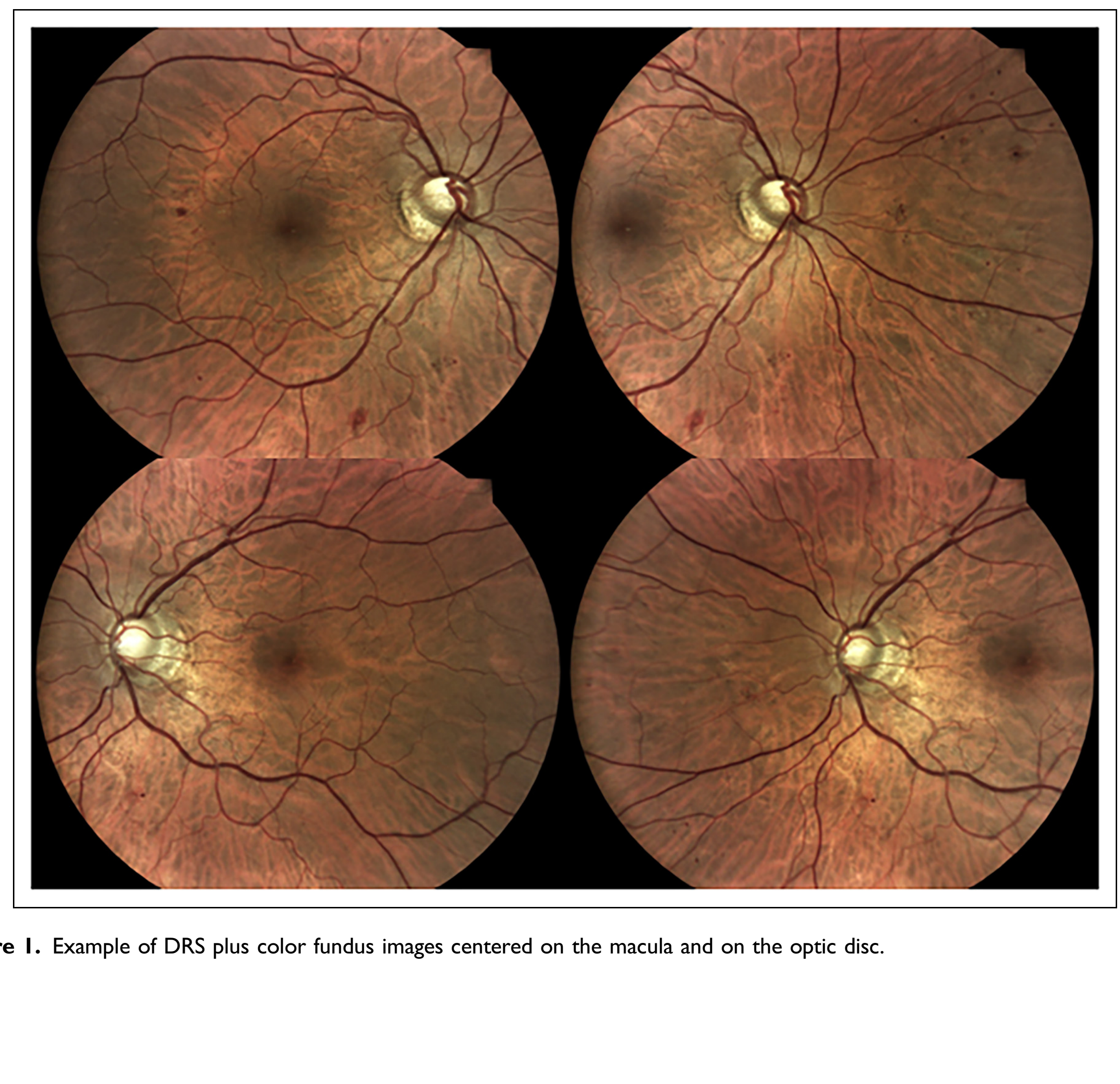

Figure 1. Example of DRS plus color fundus images centered on the macula and on the optic disc.

Participants gave their written informed consent and underwent a non-mydriatic two-field imaging acquisition protocol based on capturing two images per eye: one centred on the macula and one on the optic disc, as depicted in Figure 1. Images were collected using the iCare DRSplus fundus imaging system (Centervue, Padova, Italy).

DRSplus provides images at 45-degrees field employing a TrueColor Confocal Technology. The image capture process is quick and fully automated, hence allowing the use by non-specialized operators following minimal training. The presence of the Confocal Technology allows for good image quality even when imaging through small pupils (up to 1.8 mm as observed in this study), hence reducing screening failures in non-mydriatic conditions. DRSplus white LED illumination offers also advantages in terms of enhanced color information when compared to traditional fundus cameras, which could have implications for the detection and classification of pathological features in various eye diseases.[8] The effectiveness of True Color Confocal fundus imaging compared with standard fundus photography for DR screening has already been shown.[9]

Nurses from ASL TO5, working across two district facilities, operated the imaging system without pupil dilation. Previously, they attended a training program consisted of a specific course performed by the ophthalmologist and a two month period of field training.

In case of insufficient image quality (e.g., less than 50% of the image clearly visible), the operator could opt for a retake without pupil dilation. Only patients who continued to have ungradable images underwent to an eye examination with mydriasis.

All images were processed through iCare ILLUME, an interface for the AI-driven RetCAD software (Thirona B.V., Nijmegen, The Netherlands). Specifically, RetCAD is a class IIa CE-certified medical device software that employs AI to detect DR in colour fundus images. In addition to flagging the need for specialist referral, RetCAD classifies lesions according to the International Clinical Diabetic Retinopathy (ICDR) Disease severity Scale[10] and provides heatmaps indicating possibly abnormal areas related to DR.[11] Processing four images through RetCAD takes approximately 1 min.[12]

There is no evidence based on current available literature of any study combining iCare DRSplus and RetCAD through iCare ILLUME in Italy. RetCAD (software v2.0.2) assigned a DR severity score to each image. An eye-level DR severity score was also provided by RetCAD, derived as the highest score of the two images belonging to that eye. Moreover, a patient-level outcome was provided together with a recommendation advising referral for follow-up evaluations. The rule for patient's referral adopted by RetCAD software is DR score equal or greater than 2 (score 2, 3 or 4).The patient-level

Table 1. Characteristics of the study population.

N (%)		avg ± SD (years)	range (years)
Patients	age	68.02 ± 11.78	24-92
506 (100)	duration of diabetes	13.28 ± 11.17	0–56
Females	age	68.4 ± 12.52	24–92
216 (43)	duration of diabetes	14.09 ± 12.24	0–56
Males	age	67.73 ± 11.2	24–92
290 (57)	duration of diabetes	12.68 ± 10.28	0–48
DM1	age	46 ± 12.17	24–66
22 (4)	duration of diabetes	25.11 ± 14.9	0.5–52
DM2	age	69.09 ± 10.71	24–66
471 (93)	duration of diabetes	12.91 ± 10.76	0–56
Other types of diabetes*	age	66.31 ± 12.23	44–90
13 (3)	duration of diabetes	6.88 ± 5.08	0–14

**other types (LADA, secondary, other forms).*

Table 2. DR score distribution according to the human expert reference standard and RetCAD at eye-level and patient-level.

	Frequency			
	Human expert		RetCAD	
ICDR severity scale	Eye level N (%)	Patient level N (%)	Eye level N (%)	Patient level N (%)
No apparent retinopathy – 0	721 (71.67)	347 (68.58)	647 (64.31)	277 (54.74)
Mild nonproliferative DR – 1	126 (12.52)	76 (15.02)	143 (14.21)	103 (20.36)
Moderate nonproliferative DR – 2	61 (6.06)	33 (6.52)	90 (8.95)	52 (10.28)
Severe nonproliferative DR – 3	85 (8.45)	46 (9.09)	81 (8.05)	47 (9.29)
Proliferative DR – 4	1 (0.1)	1 (0.2)	19 (1.89)	14 (2.77)
Ungradable	12 (1.19)	3 (0.59)	26 (2.58)	13 (2.57)
Total	**1006**	**506**	**1006**	**506**

Table 3a. Confusion matrix of DR scores and ungradable cases (U) at eye-level.

EYE-LEVEL		RetCAD 0	1	2	3	4	U	Total
Ground truth (human expert)	**0**	609	90	8	0	0	14	**721**
	1	30	48	38	6	0	4	**126**
	2	1	2	31	24	0	3	**61**
	3	0	1	13	51	18	2	**85**
	4	0	0	0	0	1	0	**1**
	U	7	2	0	0	0	3	**12**
	Total	**647**	**143**	**90**	**81**	**19**	**26**	**1006**

Table 3b. Confusion matrix of referable DR (cut-off for referral set to 2) and related performance metrics (sensitivity, specificity and accuracy) at eye-level.

EYE-LEVEL		RetCAD Non-referable (0, 1)	Referable (2, 3, 4)	Total
Ground truth (human expert)	**Non-referable (0, 1)**	777	52	**829**
	Referable (2, 3, 4)	4	138	**142**
	Total	**781**	**190**	**971**

Metrics	%	95% CI
Sensitivity	97.18	2.72
Specificity	93.73	1.65
Accuracy	94.23	1.47

outcome was calculated as the maximum score of the two eyes.

All eyes were graded by a general ophthalmologist based on the ICDR severity scale. The resulting human grading is considered as the reference standard for this study. A DR score ranging from 0 (No apparent Retinopathy) to 4 (Proliferative diabetic Retinopathy) was assigned to each eye. For the purposes of this study, scores 0 and 1 (less than moderate retinopathy) were defined as non-referable retinopathy, while grades equal to 2 and above were defined as referable retinopathy (more than mild or 'sight-threatening' retinopathy).The human grading at patient-level was set as the worst DR grade of the two eyes.

The classifications provided by the human grader and RetCAD, both at eye-level and patient-level, were then compared.

The proportion of ungradable eyes out of the total captured by the DRSplus in a non-mydriatic condition was calculated. The eyes defined as ungradable by the ophthalmologist were excluded from the subsequent analysis. The ungradable judgement of the ophthalmologist reflected the impossibility to provide a DR grade based on the quality of the acquired images for that eye. RetCAD performance in identifying referable DR using the human grading as the gold standard was assessed both at eye-level and patient-level.

The statistical analysis included the following metrics: sensitivity, specificity and accuracy for the overall sample and stratified by median age. To evaluate the agreement between RetCAD and the human grading, the weighted Cohen's Kappa was calculated both at eye-level and patient-level. This is a robust method to evaluate agreement for categorical variables, allowing to take into consideration the different levels of disagreement between categories. In other words, as the DR scores deviate from perfect agreement, the weights of the disagreements linearly increase.

Results

The primary demographic characteristics of the study sample, together with diabetes type and its duration is detailed in Table 1. The average age was 68.02 ± 11.78 years (ranging from 24 to 92 years), mirroring the old population typically attending Italian diabetes centers. The frequency of males and females was 57% and 43% respectively. The sample was composed of

Table 4a. Confusion matrix of DR scores and ungradable cases (U) at patient-level.

		RetCAD						
PATIENT-LEVEL		**0**	**1**	**2**	**3**	**4**	**U**	**Total**
Ground truth (human expert)	**0**	263	70	8	0	0	6	**347**
	1	12	32	25	4	0	3	**76**
	2	1	0	15	16	0	1	**33**
	3	0	0	4	27	13	2	**46**
	4	0	0	0	0	1	0	**1**
	U	1	1	0	0	0	1	**3**
	Total	**277**	**103**	**52**	**47**	**14**	**13**	**506**

Table 4b. Confusion matrix of referable DR (cut-off for referral set to 2) and related performance metrics (sensitivity, specificity and accuracy) at patient-level.

		RetCAD		
PATIENT-LEVEL		**Non-referable (0, 1)**	**Referable (2, 3, 4)**	**Total**
Ground truth (human expert)	**Non-referable (0, 1)**	377	37	**414**
	Referable (2, 3, 4)	1	76	**77**
	Total	**378**	**118**	**491**

Metrics	%	**95% CI**
Sensitivity	98.70	2.53
Specificity	91.06	2.75
Accuracy	92.26	2.36

22(4%) patients with type 1 diabetes, 471(93%) patients with type 2 diabetes and 13 (3%) patients with other forms of diabetes. Type 1 diabetes patients were younger and with a longer duration of the disease compared to those with type 2.The prevalence of each DR score, together with the number of ungradable cases at eye-level and patient-level according to both the human expert and RetCAD is reported in Table 2. 12 eyes and 26 eyes were excluded respectively by the ophthalmologist and RetCAD because of poor quality. Thus, the ungradable rate was respectively of 1.19% (12/1006) for the human expert and of 2.58% (26/1006) for RetCAD. The dataset for ungradability analysis resulted in 994 eyes of 503 patients with an associated DR score according to the reference standard. For the final analysis to assess RetCAD performance, 971 eyes of 491 patients for which both RetCAD and human expert provided a DR score were considered. The retinopathy prevalence (grades above 0) identified by the human grader and RetCAD was respectively of 27.46% (273/994) and 33.98% (333/980).

The confusion matrix of DR grades and ungradable cases of the human expert and RetCAD is provided at eye-level in Table 3a and at patient-level in Table 4a. The weighted Cohen's kappa coefficient was of 71.87% at eye-level and 66.66% at patient-level, which corresponds to a substantial agreement[13] between RetCAD and human grader. The performance of RetCAD are reported at eye-level in Table 3b and at patient-level in Table 4b. At eye-level, RetCAD showed 97.18% (95% CI: ± 2.72%) sensitivity, 93.73% (95% CI: ± 1.65%) specificity and 94.23% (95% CI: ± 1.47%) accuracy for the detection of referable DR on DRSplus non-mydriatic retinal images. At patient-level, RetCAD exhibited 98.70% (95% CI: ± 2.53%) sensitivity, 91.06% (95% CI: ± 2.75%)specificity and 92.26% (95% CI: ± 2.36%) accuracy for the detection of referable DR.

The same analysis was performed for two groups resulting from the stratification of the study sample according to median age (70 years): group 1 with age below the median, and group 2 with age above the median. The performance for group 1 is reported at eye-level in Table 5a and at patient-level in Table 6a, for group 2 are reported at eye-level in Table 5b and at patient-level in Table 6b. An ungradability analysis was performed to evaluate the ungradable rates at both eye-level and patient-level for the overall study sample and the two groups stratified by age (Table 7).

Discussion and conclusions

Fundus imaging is an established methodology to detect DR.[14,15] Digital retinal photography has revolutionized

Table 5a. Confusion matrix at eye-level for patients below the median age of the sample: DR grades and ungradable (U) cases (left); referable DR and performance metrics (right).

Group 1 (age ≤ 70 years

EYE-LEVEL		RetCAD 0	1	2	3	4	U	Total
Ground truth (human expert)	0	355	24	0	0	0	3	**382**
	1	10	23	24	1	0	3	**61**
	2	1	0	19	16	0	2	**38**
	3	0	1	9	23	12	1	**46**
	4	0	0	0	0	1	0	**1**
	U	2	0	0	0	0	0	**2**
	Total	**368**	**48**	**52**	**40**	**13**	**9**	**530**

EYE-LEVEL		RetCAD Non-referable (0, 1)	Referable (2, 3, 4)	Total
Ground truth (human expert)	Non-referable (0, 1)	412	25	**437**
	Referable (2, 3, 4)	2	80	**82**
	Total	**414**	**105**	**519**

Metrics	%	95% CI
Sensitivity	97.56	3.34
Specificity	94.28	2.18
Accuracy	94.80	1.91

Table 5b. Confusion matrix at eye-level for patients above the median age of the sample: DR grades and ungradable (U) cases (left); referable DR and performance metrics (right).

Group 2 (age > 70 years)

EYE-LEVEL		RetCAD 0	1	2	3	4	U	Total
Ground truth (human expert)	0	254	66	8	0	0	11	**339**
	1	20	25	14	5	0	1	**65**
	2	0	2	12	8	0	1	**23**
	3	0	0	4	28	6	1	**39**
	4	0	0	0	0	0	0	**0**
	U	5	2	0	0	0	3	**10**
	Total	**279**	**95**	**38**	**41**	**6**	**17**	**476**

EYE-LEVEL		RetCAD Non-referable (0, 1)	Referable (2, 3, 4)	Total
Ground truth (human expert)	Non-referable (0, 1)	365	27	**392**
	Referable (2, 3, 4)	2	58	**60**
	Total	**367**	**85**	**452**

Metrics	%	95% CI
Sensitivity	96.67	4.54
Specificity	93.11	2.51
Accuracy	93.58	2.26

DR screening by offering quick examination times, high image quality, and simple image storage and management.[16,17] Such advancements made large-scale screening programs feasible.[18,19]

In the last decades, automated retinal image analysis software (ARIAS) to detect DR began to emerge.[20] ARIAS adoption in screening programs could make the process more efficient, cost-effective, reproducible and accessible, handling a large and growing diabetic population. In addition, ARIAS could also improve manual DR screening, which is a time-consuming activity affected by both inter- and intra-grader variability.[21]

This observational cross-sectional study aimed to evaluate the screening system combining DRSplus confocal fundus camera and RetCAD AI software through iCare ILLUME in terms of both image quality and performance in DR grading. The DR scores of RetCAD were compared with those of an expert ophthalmologist, who represented the reference standard (ground truth). During this study, such screening system was employed for the first time in Italian diabetology facilities, facing the challenges of collecting high quality fundus images in non-mydriatic conditions and achieving a high DR screening success rate.

The ungradable rate of the ophthalmologist was very low (around 1.2%), almost half with respect to the 2.6% observed for RetCAD. This result can be attributed to the subjectivity of the quality evaluation performed by one

Table 6a. Confusion matrix at patient-level for patients below the median age of the sample: DR grades and ungradable (U) cases (left); referable DR and performance metrics (right).

Group 1 (age ≤ 70 years)

PATIENT-LEVEL		RetCAD 0	1	2	3	4	U	Total
Ground truth (human expert)	**0**	164	21	0	0	0	0	**185**
	1	5	13	14	1	0	2	**35**
	2	1	0	8	9	0	1	**19**
	3	0	0	3	14	8	1	**26**
	4	0	0	0	0	1	0	**1**
	U	0	0	0	0	0	0	**0**
	Total	**170**	**34**	**25**	**24**	**9**	**4**	**266**

PATIENT-LEVEL		RetCAD Non-referable (0, 1)	Referable (2, 3, 4)	Total
Ground truth (human expert)	**Non-ref (0, 1)**	203	15	**218**
	Ref (2, 3, 4)	1	43	**44**
	Total	**204**	**58**	**262**

Metrics	%	95% CI
Sensitivity	97.73	4.40
Specificity	93.12	3.36
Accuracy	93.89	2.90

Table 6b. Confusion matrix at patient-level for patients above the median age of the sample: DR grades and ungradable (U) cases (left); referable DR and performance metrics (right).

Group 2 (age > 70 years)

PATIENT-LEVEL		RetCAD 0	1	2	3	4	U	Total
Ground truth (human expert)	**0**	99	49	8	0	0	6	**162**
	1	7	19	11	3	0	1	**41**
	2	0	0	7	7	0	0	**14**
	3	0	0	1	13	5	1	**20**
	4	0	0	0	0	0	0	**0**
	U	1	1	0	0	0	1	**3**
	Total	**107**	**69**	**27**	**23**	**5**	**9**	**240**

PATIENT-LEVEL		RetCAD Non-referable (0, 1)	Referable (2, 3, 4)	Total
Ground truth (human expert)	**Non-ref (0, 1)**	174	22	**196**
	Ref (2, 3, 4)	0	33	**33**
	Total	**174**	**55**	**229**

Metrics	%	95% CI
Sensitivity	100.00	0.00
Specificity	88.78	4.42
Accuracy	90.39	3.82

Table 7. Ungradability analysis at eye-level and patient-level for the overall study sample and stratified by median age.

		Ungradability analysis	Age stratification analysis	
Level analysis	**Grader**	**Total sample analysis N/total (%)**	**Group 1 (age ≤ 70 years) N group 1/total ungradable (%)**	**Group 2 (age > 70 years) N group 2/total ungradable (%)**
Eye-level	RetCAD	26/1006 (2.58%)	9/26 (34.62%)	17/26 (65.38%)
	Human expert	12/1006 (1.19%)	2/12 (16.67%)	10/12 (83.33%)
Patient-level	RetCAD	13/506 (2.57%)	4/13 (30.77%)	9/13 (69.23%)
	Human expert	3/506 (0.59%)	0/3 (0.00%)	3/3 (100%)

human expert, whose criteria for ungradability may differ from those of RetCAD. In any case, the ungradability rate observed in this study for RetCAD is in line with the values reported in literature.[20,22,23] Such rate can be considered more than reasonable in a population of patients with advanced age attending a diabetology clinic for DR screening. In our study, the majority of ungradable cases were observed in the older group of patients with age above median for both RetCAD (around 65% at eye-level and 69% at patient-level) and the human expert (around 83% at eye-level and 100% at patient-level). These results can be explained considering that older patients are often affected by cataracts and small pupils, which are known factors affecting fundus image quality.[23] Overall, the very low ungradability rates of both RetCAD and the human grader emphasizes the ability of DRSplus to obtain high-quality images, even in non-mydriatic conditions. Moreover, RetCAD quality score allowing to automatically differentiate between a poor and a sufficient image quality, may be very helpful in diabetology facilities where the AI software is used by non-specialized operators and ophthalmologists are not involved in the first screening phases.[22]

The performance of this system, evaluated in a real-world clinical scenario, showed both at eye-level and patient-level impressive sensitivity and specificity values, respectively above 97% and 91%. From a practical point of view, a sensitivity value of 97% means that only 3 patients out of 100 would be incorrectly identified as referable for DR. On the other hand, a specificity of 91% means that 9 patients without the need of a further DR evaluation would be unnecessarily referred to an ophthalmologist. These results are in line with the performance achieved by automated grading systems already reported in literature[20–22]). For perspective, the first AI diagnostic system to receive FDA approval achieved a sensitivity of 87.2% and a specificity of 90.7%.[18,24] Additionally, IDX-DR (Idx LLC, Iowa, USA) recorded 91% sensitivity and 84% specificity,[25] EYEART app reported 91.7% sensitivity and 91.5% specificity,[25,26] and RETMARKER app showed 85% sensitivity.[27] Moreover, as required by a robust AI-based screening system to detect DR, the performance was consistent across the two sub-groups of patients stratified by median age.

Theoretically, an AI-based system intended for screening a potentially sight-threatening pathology such as DR, would benefit from a high sensitivity value in the trade-off between sensitivity and specificity.[17] As a matter of fact, a high sensitivity values ideal in the context of AI-based screening since it minimizes the risk of obtaining false negative outcomes and of misclassifying patients needing a follow-up evaluation by an ophthalmologist. The system employed in this study shows performance perfectly in line with these expectations. These results suggested that our AI-based screening system can potentially reduce the number of undetected DR patients, leading to earlier interventions and better patient outcomes. On the other hand, the system showed a specificity above 91% that can potentially mitigate unnecessary referrals to eye-care specialists and support efficient healthcare resources allocation.

The outcomes of this research suggest that the proposed AI-based screening solution combining DRSplus for image acquisition and RetCAD software for image analysis can be integrated into clinical daily practice by simplifying and making the screening process more efficient for both patients and eye-care providers. Similarly, this study emphasizes the importance of developing streamlined, cost-effective approaches for retinopathy screening in expanding populations such as those with diabetes. Minimizing the screening workload on ophthalmologists is key, advocating for automated image evaluation systems and reserving the more specialized task of treatment for experts. The primary limitation of the current study is the gold standard grading being based on a single human expert evaluation rather than on more independent human graders. This implies that the ground truth used in this study may be strengthened by increasing the number of graders to reduce the chances of grading errors (e.g., from fatigue or missed subtle findings) and to enable the evaluation of the inter-grader agreement. Nonetheless, the strength of this study lies in its real-world approach adopted for DR screening, executed by a diabetes healthcare team with a minimum training for fundus image acquisition. Such setting is common and resembles a real-world scenario in which patients afferent to diabetology facilities are screened and referred for DR to the ophthalmologist only if needed. The screening system described in this study, involving a TrueColor confocal fundus imaging system combined with RetCAD AI software, can provide real-time DR detection in diabetology facilities with performances similar to those of eye-care specialists.

Authorship contribution

Piatti Alberto: Conceptualization, Methodology, Writing- Original draft preparation and Supervision. **Rui Chiara:**Data curation, Methodology, Writing- Original draft preparation and Supervision.**Gazzina Silvia:** Data curation, Writing- Reviewing and Editing. **Tartaglino Barbara:** Data curation, Writing- Reviewing and Editing**. Romeo Francesco:** Data curation, Writing- Reviewing and Editing. **Manti Roberta:** Data curation, Writing- Reviewing and Editing. **DoglioMarella:** Data curation, Writing- Reviewing and Editing. **Nada Elisa:** Data curation, Writing- Reviewing and Editing. **Giorda Carlo Bruno:** Conceptualization, Methodology, Writing- Original draft preparation and Supervision.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Disclosure and Conflict of Interest

Piatti A is consultant of Icare.

Tartaglino B, Manti R, Romeo F, Doglio M, Nada E and Giorda CB declare that they have no conflicts of interest.

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